

MINGXUAN LI

Causal Artificial Intelligence Lab, Columbia University, New York, NY 10027

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EDUCATION

Columbia University

Sept. 2021 - Now

Ph.D. Candidate in Computer Science

Graduate Research Assistant at Columbia University (09/2021-Present)

Advisor: Elias Bareinboim

Brown University

Sept. 2019 - June 2021

Sc.M. in Computer Science, GPA: 4.0/4.0

Advisor: Michael L. Littman

Beihang University

Sept. 2015 - June 2019

B.Sc. in Computer Science and Technology, GPA: 3.7/4.0

Advisor: Jingyuan Wang

RESEARCH INTERESTS

Causal Reinforcement Learning, Embodied AI, Agentic System, Alignment

SELECTED PUBLICATIONS

- “Counterfactual Debugging the World Model Transfer Gap”
NeurIPS 2026 - **Mingxuan Li***, Kai-Zhan Lee*, Michael Dennis, Elias Bareinboim
- “Counterfactual Shapley Credit Assignment”
RLC 2026 - **Mingxuan Li***, Kai-Zhan Lee*, Elias Bareinboim
- “Scalable Causal Imitation Learning”
RLC 2026 - Eylam Tagor, **Mingxuan Li**, Elias Bareinboim
- “Causal Flow Q-Learning for Robust Offline Reinforcement Learning”
ICML 2026 - **Mingxuan Li**, Junzhe Zhang, Elias Bareinboim
- “Confounding Robust Continuous Control via Automatic Reward Shaping”
AAMAS 2026 (Oral) - Mateo Juliani*, **Mingxuan Li***, Elias Bareinboim
- “Confounding Robust Deep Reinforcement Learning: A Causal Approach”
NeurIPS 2025 - **Mingxuan Li***, Junzhe Zhang*, Elias Bareinboim
- “Automatic Reward Sharing from Confounded Offline Data”
ICML 2025 - **Mingxuan Li**, Junzhe Zhang, Elias Bareinboim
- “Causally Aligned Curriculum Learning”
ICLR 2024 - **Mingxuan Li**, Junzhe Zhang, Elias Bareinboim
- “Learning Generalizable Behavior via Visual Rewrite Rules”
AAAI 2022 Workshop on Reinforcement Learning in Games - Yiheng Xie*, **Mingxuan Li***, Shangqun Yu*, Michael L. Littman
- “Interpretability is a Kind of Safety: An Interpreter-based Ensemble for Adversary Defense”
KDD 2020 (Oral) - Jingyuan Wang, Yufan Wu, **Mingxuan Li**, Xin Lin, Junjie Wu, Chao Li

INDUSTRIAL EXPERIENCE

Amazon

May 2026 - Aug. 2026

Advisor: Dr. Josh Hooks

Applied Scientist Intern, Amazon Robotics, Vulcan Stow

- Built the full dataset preparation to large scale distributed reinforcement learning training pipeline from scratch;
- Proposed a transformer based offline reinforcement learning agent with online tuning capability that optimizes stowing decisions continuously;
- Improved stowing speed by 3% and success rate by 4pt compared to the production baseline.

Uber

Jun. 2025 - Aug. 2025

Advisor: Jeffery Shen

Research Scientist Intern, Uber San Francisco

- Conducted large-scale user tipping behavior analysis and constructed a sequential user behavior dataset on billions of logs with Spark;
- Prototyped and helped productionize Epidemic Neural Recommenders (ENR) to enhance the non-linear representation learning capability of Uber's contextual bandit platform for usage across the company;
- Prototyped and helped productionize large-scale customized Decision Transformers (DT) to model sequential user behaviors and to automate time series feature engineering; Beat the current SOTA user behavior prediction model on internal datasets by over 30%.

Amazon

Jun. 2021 - Aug. 2021

Advisor: Dr. Prag Mishra

Applied Scientist Intern, Amazon RAI

- Analyzed billions of delivery trajectories to identify the bottleneck in delivery route planning;
- Proposed an online reinforcement learning agent that tunes the delivery route planning algorithm's hyperparameters automatically based on recent performance;
- Improved delivery efficiency by 15% compared to the previous algorithm.

TEACHING

- Lecturer, CS 4775W, Causal Inference II, Columbia University, Spring/2025.
- TA, CS 6998, Causal Trustworthy AI, Columbia University, Fall/2023.
- TA, DATA 2040, Deep Learning and Special Topics in Data Science, Brown University, Spring/2020.

COMMUNITY SERVICES

- Organizer: Causal Reinforcement Learning Workshop, RLC, 2025
- Conference Reviewer: ICML, 2025-2026; NeurIPS, 2023-2026; ICLR, 2024-2026; AISTATS, 2025-2026; AAAI, 2026.
- Journal Reviewer: Journal of Machine Learning Research (JMLR); International Journal of Robotics Research (IJRR); Transactions on Pattern Analysis and Machine Intelligence (TPAMI); Transactions on Machine Learning Research (TMLR).

SKILLS

Computer Languages
Software
Language

Python, Java, C/C++, SQL, HTML
L^AT_EX, Pytorch, Jax, TensorFlow
Chinese(Native), English(Fluent)

AWARDS&HONOURS

01/2026, MATS Scholarship Cohort 9.0 for AI Alignment & Safety (Global Top 7%)

12/2025, Top Reviewer @ NeurIPS 2025 (Top 8%)

12/2025, Travel Award @ NeurIPS 2025

10/2018, Scholarship for Academic Achievements, Second Prize (Top 10%)

09/2018, Was selected to appear on the Dean's List for the School of Engineering, HKUST

09/2017, The 1st National Student Computer System Capability Challenge, Second Prize (Final 2/70)

05/2017, The 27th "FengRu Cup" University Students Extra-Curricular Scientific and Technological Invention Competition, Second Prize (Final 4/176)